

第六次习题课 Section 12.1-12.6

考點:

$$1. C. \quad |\vec{a} + \vec{b}| = \sqrt{|\vec{a}|^2 + |\vec{b}|^2} = |\vec{a} - \vec{b}|$$

$$\vec{a} \cdot \vec{b} = 0$$

$$2. T. \quad \vec{a} \times \vec{b} = \vec{a} \times \vec{c} \Rightarrow \vec{a} \times (\vec{b} - \vec{c}) = 0 \Rightarrow \vec{a} - \vec{c} = \vec{0}$$

$$\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c} \Rightarrow \vec{a} \cdot (\vec{b} - \vec{c}) = 0 \quad \vec{a} = \vec{c}$$

$$3. \quad 0 = (\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} + \vec{b} + \vec{c}) = \underbrace{|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2}_{3} + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$$

$$\Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = \underline{\underline{-\frac{3}{2}}}$$

$$4. \quad \vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 1 \\ -1 & 1 & 1 \end{vmatrix} = \vec{i} - 2\vec{j} + 3\vec{k}$$

$$\Rightarrow \vec{c} = k(\vec{i} - 2\vec{j} + 3\vec{k})$$

$$\vec{c} \cdot (\vec{i} - 2\vec{j} + \vec{k}) = 16 \Rightarrow k \cdot (1 + 4 + 3) = 16 \Rightarrow k = 2$$

$$\Rightarrow \vec{c} = (2, -4, 6)$$

$$\text{考點: } \vec{u} = (2, 1, -1) \Rightarrow \vec{n} = \vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & -1 \\ -2 & 1 & -1 \end{vmatrix} = 4\vec{j} + 4\vec{k}$$

$$\vec{v} = (-2, 1, -1)$$

$$P(-1, 3, 0)$$

$$\Rightarrow 4y - 3 + 4z = 0 \Rightarrow \underline{\underline{y + z = \frac{3}{4}}}$$

Record good times

2. $\vec{v}_1 = (1, 0, 1)$

$\vec{v}_2 = (2, 0, 2)$

$$\vec{n} = \vec{v}_1 \times \vec{v}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 1 \\ 2 & 0 & 2 \end{vmatrix} = \vec{0}$$

$\Rightarrow \vec{v}_1 \parallel \vec{v}_2 \Rightarrow L_1 \parallel L_2 \Rightarrow L_1 \text{ and } L_2 \text{ are in one plane}$

$L_1: P(0, 1, 1)$

$\Rightarrow \vec{p\vec{a}} = (1, -1, -1)$

$$\Rightarrow \vec{n} = \vec{v}_1 \times \vec{p\vec{a}} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 1 \\ 1 & -1 & -1 \end{vmatrix} = \vec{i} + 2\vec{j} - \vec{k}$$

$L_2: Q(1, 0, 0)$

$\Rightarrow \text{the plane: } x + 2(y-1) - (z-1) = 0 \Rightarrow x + 2y - z = 1$

3. $\vec{v}_1 = (0, 1, 1)$

$\vec{v}_2 = (1, 2, 1)$

$$\Rightarrow \vec{n} = \vec{v}_1 \times \vec{v}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{vmatrix} = -\vec{i} + \vec{j} - \vec{k}$$

$P(0, 0, 0)$

$\Rightarrow \text{the plane: } 0x - y + z = 0$

4. $\vec{n}_1 = (1, -2, 3)$

$\vec{n}_2 = (1, 2, -3)$

$$\Rightarrow \vec{n} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -2 & 3 \\ 1 & 2 & -3 \end{vmatrix} = 6\vec{j} + 4\vec{k}$$

$P(1, 0, 1)$

$\Rightarrow \text{the plane: } 6y + 4(z-1) = 0 \Rightarrow 3y + 2z = 2$

5. $P(2, -1, -1), Q(1, 0, -1) \Rightarrow \vec{p\vec{a}} = (-1, 1, 0), \vec{n}_1 = (2, 3, -5)$

$$\vec{n} = \vec{p\vec{a}} \times \vec{n}_1 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 1 & 0 \\ 2 & 3 & -5 \end{vmatrix} = -5\vec{i} - 5\vec{j} - 5\vec{k}$$

$\Rightarrow \text{the plane:}$

$(x-2) + (y+1) + (z+1) = 0$
 $x + y + z = 0$

考点三: D

$$\vec{v}_1 = (1, 2, -1) \Rightarrow \vec{v}_1 \cdot \vec{v}_2 = -1$$

$$\vec{v}_2 = (2, 1, 1)$$

$$\text{if it intersect: } \begin{cases} x=t=1-2t \\ y=2t=t \end{cases} \Rightarrow \begin{cases} t=\frac{1}{3} \\ t=0 \end{cases} \times$$

2. D.

$$\vec{n}_1 = (1, 2, -1) \Rightarrow \cos \theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| |\vec{n}_2|} = \frac{3}{\sqrt{6} \cdot \sqrt{2}} = \frac{3}{2\sqrt{3}} = \frac{\sqrt{3}}{2}$$

$$\vec{n}_2 = (1, 1, 0)$$

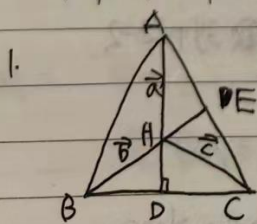
$$\Rightarrow \theta = \frac{\pi}{6}$$

$$3. \text{ T. } \vec{n}_1 = (1, 1, -2) \quad \vec{n}_2 = (1, 1, 1) \Rightarrow \vec{n}_1 \cdot \vec{n}_2 = 0 \Rightarrow \vec{n}_1 \perp \vec{n}_2$$

$$4. \quad S: (1, 1, 1) \quad P: (0, -2, 0) \Rightarrow \vec{PS} = (1, 3, 1), \quad \vec{n} = (1, 2, 3)$$

$$\Rightarrow d = \left| \frac{\vec{PS} \cdot \vec{n}}{|\vec{n}|} \right| = \frac{|1+6+3|}{\sqrt{14}} = \frac{10}{\sqrt{14}}$$

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$$\text{let } \vec{AH} = \vec{a}, \quad \vec{BE} = \vec{b}, \quad \vec{AC} = \vec{c}$$

$$\begin{aligned} \Rightarrow \vec{a} \cdot \vec{BC} = 0 &\Rightarrow \begin{cases} \vec{a} \cdot (\vec{c} - \vec{b}) = 0 \\ \vec{b} \cdot \vec{AC} = 0 \end{cases} \Rightarrow \begin{cases} \vec{a} \cdot (\vec{c} - \vec{b}) = 0 \\ \vec{b} \cdot (\vec{a} - \vec{c}) = 0 \end{cases} \\ \Rightarrow \vec{c} \cdot (\vec{a} - \vec{b}) = 0 &\Rightarrow \vec{c} \cdot \vec{AB} = 0 \\ \Rightarrow CH \perp AB \end{aligned}$$

$$\begin{aligned}
 9. \quad & O(0,0,0) \Rightarrow \vec{OP} = (6, -3, 2) \Rightarrow \vec{n} = \vec{OP} \times \vec{n}_1 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 6 & -3 & 2 \\ 4 & -1 & 2 \end{vmatrix} \\
 & P(6, -3, 2) \quad \vec{n}_1 = (4, -1, 2) \\
 & \quad \quad \quad = -4\vec{i} - 4\vec{j} + 6\vec{k}
 \end{aligned}$$

$$\Rightarrow \text{the plane: } 2x + 2y - 3z = 0$$

$$11. \quad \sqrt{(x-3)^2 + (y-1)^2 + (z-2)^2} = \sqrt{(x-4)^2 + (y+2)^2 + (z+2)^2} = \sqrt{(x+1)^2 + (y+5)^2 + (z-1)^2}$$

$$\begin{aligned}
 \Rightarrow \begin{cases} x-3y-4z=5 \\ 3x-4y+z=-6 \end{cases} \quad \vec{n}_1 = (1, 3, -4) \quad \vec{n}_2 = (3, -4, 1) \Rightarrow \vec{n} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 3 & -4 \\ 3 & -4 & 1 \end{vmatrix} \\
 \quad \quad \quad = -19\vec{i} - 13\vec{j} + 5\vec{k}
 \end{aligned}$$

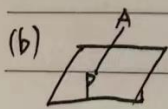
$$P(0, 1, -2) \Rightarrow \begin{cases} x = -19t \\ y = 1 - 13t \\ z = -2 + 5t \end{cases}$$

即过三点的直线，且垂直于这三点所构成的平面的直线

$$\begin{aligned}
 18. (a) \quad & \vec{n}_1 = (1, -1, 1) \Rightarrow \vec{n} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 1 \\ 3 & -1 & 1 \end{vmatrix} = 2\vec{j} + 2\vec{k} \\
 & \vec{n}_2 = (1, 5, 6)
 \end{aligned}$$

$$P(1, -1, 0)$$

$$\Rightarrow \text{the plane: } 2(y+1) + 2z = 0 \Rightarrow y+z = -1$$



$$\begin{aligned}
 \begin{cases} x = 1+t \\ y = -1-t \\ z = t \end{cases} \Rightarrow t = \frac{1}{5} \Rightarrow P\left(\frac{6}{5}, -\frac{6}{5}, \frac{1}{5}\right) \\
 3x - 4y + 2z = 5
 \end{aligned}$$

$$\vec{v} = \vec{n} \times \vec{n} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 2 & 2 \\ 3 & -1 & 1 \end{vmatrix} = 4\vec{i} + 6\vec{j} - 6\vec{k}$$

$$\Rightarrow \begin{cases} x = 4t + \frac{6}{5} \\ y = 6t - \frac{6}{5} \\ z = -6t + \frac{1}{5} \end{cases}$$

19. $L: \begin{cases} x+5y+6z=3 \\ x-2y+3z=1 \end{cases}$ xOy 平面: $z=0$

$$\Rightarrow L \text{ 与 } xOy \text{ 面交点: } \begin{cases} x+5y=3 \\ x-2y=1 \end{cases} \Rightarrow \begin{cases} x = \frac{11}{7} \\ y = \frac{2}{7} \\ z = 0 \end{cases} \quad A(\frac{11}{7}, \frac{2}{7}, 0)$$

L 上任取点 $P(1, 0, \frac{2}{3}) \Rightarrow$ 点 P 在平面 xOy 的投影点 $B(1, 0, 0)$

$$\Rightarrow \vec{v} = \vec{BA} = (\frac{18}{7}, \frac{2}{7}, 0) \Rightarrow \text{投影} \begin{cases} x = \frac{11}{7} + t \\ y = \frac{2}{7} + t \\ z = 0 \end{cases}$$

$$A(\frac{11}{7}, \frac{2}{7}, 0)$$